

Taichi MLS-MPM Dough Model

1 State Variables

The simulation represents dough with material particles coupled to a background grid. For each particle p the state is

$$\mathbf{x}_p, \quad \mathbf{v}_p, \quad \mathbf{F}_p, \quad \mathbf{C}_p, \quad J_{p,p}.$$

Here $\mathbf{x}_p$ is position, $\mathbf{v}_p$ is velocity, $\mathbf{F}_p$ is the deformation gradient, $\mathbf{C}_p$ is the APIC affine velocity matrix, and $J_{p,p}$ is the accumulated plastic volume ratio. Grid nodes are indexed by i and store

$$\mathbf{v}_i, \quad m_i.$$

2 Particle Initialization

Particles are sampled inside an ellipsoid centered at

$$\mathbf{x}_0 = (0.5, 0.32, 0.5)^T$$

with radii

$$\mathbf{r} = (0.14, 0.055, 0.12)^T.$$

For a random point $\mathbf{q}$ inside the unit sphere,

$$\mathbf{x}_p = \mathbf{x}_0 + \mathbf{r} \odot \mathbf{q}, \quad \mathbf{v}_p = \mathbf{0}, \quad \mathbf{F}_p = \mathbf{I}, \quad \mathbf{C}_p = \mathbf{0}, \quad J_{p,p} = 1.$$

3 Material Parameters

The Lamé parameters are computed from Young's modulus E and Poisson ratio ν :

$$\mu_0 = \frac{E}{2(1+\nu)}, \quad \lambda_0 = \frac{E\nu}{(1+\nu)(1-2\nu)}.$$

If plastic volume hardening is enabled, stiffness is modified by

$$h_p = \exp(h_J(1 - J_{p,p})), \quad \mu = h_p\mu_0, \quad \lambda = h_p\lambda_0,$$

where h_J is the **jp-hardening** coefficient. With $h_J = 0$, $J_{p,p}$ is tracked but does not change the stiffness.

4 MLS-MPM Transfer

Let the grid spacing be Δx and $\Delta x^{-1} = n_{\text{grid}}$. For particle p ,

$$\mathbf{g}_p = \mathbf{x}_p / \Delta x, \quad \mathbf{b}_p = \lfloor \mathbf{g}_p - 0.5 \rfloor, \quad \mathbf{f}_p = \mathbf{g}_p - \mathbf{b}_p.$$

The one-dimensional quadratic B-spline weights are

$$w_0(f) = \frac{1}{2}(1.5 - f)^2, \quad w_1(f) = 0.75 - (f - 1)^2, \quad w_2(f) = \frac{1}{2}(f - 0.5)^2.$$

For a neighboring node offset $\mathbf{o} = (a, b, c)$, $a, b, c \in \{0, 1, 2\}$,

$$W_{p\mathbf{o}} = w_a(f_x)w_b(f_y)w_c(f_z), \quad \mathbf{d}_{p\mathbf{o}} = (\mathbf{o} - \mathbf{f}_p)\Delta x.$$

5 Deformation Gradient Update

Before particle-to-grid transfer, the deformation gradient is updated by the local velocity gradient:

$$\mathbf{F}_p^* = (\mathbf{I} + \Delta t \mathbf{C}_p) \mathbf{F}_p^n.$$

6 SVD Clamp Plasticity

If plasticity is enabled, the model performs singular value clamping:

$$\mathbf{F}_p^* = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T, \quad \tilde{\sigma}_k = \text{clamp}(\sigma_k, \sigma_{\min}, \sigma_{\max}).$$

The projected deformation gradient is

$$\mathbf{F}_p^{n+1} = \mathbf{U} \tilde{\mathbf{\Sigma}} \mathbf{V}^T.$$

In the implementation,

$$\sigma_{\min} = \text{plastic-min}, \quad \sigma_{\max} = \text{plastic-max}.$$

Values outside this interval are treated as plastic deformation, so only the clamped part remains elastic.

7 Plastic Volume Ratio

If J_p tracking is enabled, the accumulated plastic volume ratio is updated as

$$J_{p,p}^{n+1} = \text{clamp} \left(J_{p,p}^n \frac{\det(\mathbf{F}_p^*)}{\det(\mathbf{F}_p^{n+1})}, J_{p,\min}, J_{p,\max} \right).$$

Here $J_{p,\min}$ and $J_{p,\max}$ correspond to `jp-min` and `jp-max`.

8 Constitutive Model

The model uses corotated elastic stress plus a simple viscous term. Let

$$J = \det(\mathbf{F}_p), \quad \mathbf{F}_p = \mathbf{R} \mathbf{S}$$

be the polar decomposition. The elastic stress term used in the code is

$$\mathbf{P}_e = 2\mu(\mathbf{F}_p - \mathbf{R})\mathbf{F}_p^T + \lambda J(J - 1)\mathbf{I}.$$

The viscous term is

$$\mathbf{P}_v = \eta(\mathbf{C}_p + \mathbf{C}_p^T),$$

where η is `viscosity`. The total stress-like quantity used for MLS-MPM transfer is

$$\mathbf{P} = \mathbf{P}_e + \mathbf{P}_v.$$

9 Particle-to-Grid Transfer

The affine momentum contribution is

$$\mathbf{A}_p = -\Delta t V_p 4 \Delta x^{-2} \mathbf{P} + m_p \mathbf{C}_p.$$

Grid mass and momentum are accumulated as

$$\begin{aligned} m_i &+= W_{pi} m_p, \\ m_i \mathbf{v}_i &+= W_{pi} (m_p \mathbf{v}_p + \mathbf{A}_p \mathbf{d}_{pi}). \end{aligned}$$

After accumulation,

$$\mathbf{v}_i \leftarrow \frac{\mathbf{v}_i}{m_i}.$$

10 Grid Update

Gravity is applied on grid nodes:

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t (0, g, 0)^T.$$

The floor is a plane at $y = y_f$. If a grid node is below the floor and moving downward,

$$v_{i,y} \leftarrow 0, \quad v_{i,x} \leftarrow 0.7 v_{i,x}, \quad v_{i,z} \leftarrow 0.7 v_{i,z}.$$

11 Tool Box Contact

Each tool is an oriented box with center $\mathbf{c}$, rotation $\mathbf{Q}$, visible half-size

$$\mathbf{h} = (0.05, 0.05, 0.05)^T,$$

and collision half-size

$$\mathbf{h}_c = \mathbf{h} + \epsilon_c,$$

where ϵ_c is `tool-contact-padding`. A point $\mathbf{x}$ is converted to tool-local coordinates by

$$\mathbf{q} = \mathbf{Q}^T (\mathbf{x} - \mathbf{c}).$$

Collision is active when

$$|q_x| < h_{c,x}, \quad |q_y| < h_{c,y}, \quad |q_z| < h_{c,z}.$$

The contact normal is chosen from the smallest penetration axis. For relative velocity

$$\mathbf{u} = \mathbf{v} - \mathbf{v}_{tool},$$

the inward normal component is removed:

$$\mathbf{u} \leftarrow \begin{cases} \mathbf{u} - (\mathbf{u} \cdot \mathbf{n}) \mathbf{n}, & \mathbf{u} \cdot \mathbf{n} < 0, \\ \mathbf{u}, & \text{otherwise.} \end{cases}$$

The remaining relative velocity is damped by friction and absorption:

$$\mathbf{v} \leftarrow \mathbf{v}_{tool} + \alpha_f (1 - \alpha_a) \mathbf{u},$$

where α_f is `tool-contact-friction` and α_a is `tool-contact-absorption`. The same idea is applied on grid nodes and again as a particle-level projection to reduce tunneling.

12 Grid-to-Particle Transfer

Particle velocity and affine velocity matrix are interpolated back from the grid:

$$\mathbf{v}_p^{n+1} = \sum_i W_{pi} \mathbf{v}_i,$$

$$\mathbf{C}_p^{n+1} = 4\Delta x^{-1} \sum_i W_{pi} \mathbf{v}_i \mathbf{d}_{pi}^T.$$

Global numerical damping is then applied:

$$\mathbf{v}_p^{n+1} \leftarrow \gamma_v \mathbf{v}_p^{n+1},$$

where γ_v is **velocity-damping**. Finally,

$$\mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \Delta t \mathbf{v}_p^{n+1}.$$

13 Additional Plastic Damping

If a particle yielded during SVD clamping, additional damping is applied:

$$\mathbf{v}_p \leftarrow \gamma_p \mathbf{v}_p, \quad \mathbf{C}_p \leftarrow \gamma_C \mathbf{C}_p,$$

where γ_p is **plastic-velocity-damping** and γ_C is **plastic-affine-damping**. This is a pragmatic numerical term that reduces rubber-like rebound after plastic deformation.

14 Interpretation

The current material should be described as a simplified MLS-MPM dough-like model:

corotated elasticity + viscous stress + SVD clamp plasticity + optional J_p hardening.

It is inspired by common snow/plastic MPM formulations, but it is not yet a calibrated food-dough rheology model.